

1. INTRODUCTION

The objective of this project was to design and develop an Advanced Library Management System utilizing Object-Oriented Programming principles, C# Windows Forms, and a Microsoft SQL Server database. To ensure high maintainability, security, and scalability, the system was built using a strict 3-Tier Architecture, separating the Data Access Layer (DAL), Business Logic Layer (BLL), and User Interface (UI).

The application provides a seamless, dynamic user experience with features such as role-based authentication, real-time database grid synchronization, and a custom "Wishlist" logic gate. By utilizing parameterized SQL queries, the system is protected against SQL injection attacks, and the UI incorporates double-buffering techniques to ensure buttery-smooth navigation across massive datasets.

2. FEATURE LIST

The system is designed around role-based access control, ensuring that different user types have distinct permissions and interfaces.

2.1 User 1: Management (Admin)

- **Secure Authentication:** Access to the management dashboard is strictly gated by role verification.
- **Inventory Control:** Ability to add new books to the global library database.
- **Inventory Pruning:** Ability to permanently remove books from the library system.
- **Live Dashboard:** A real-time data grid displaying all current books, authors, and availability.

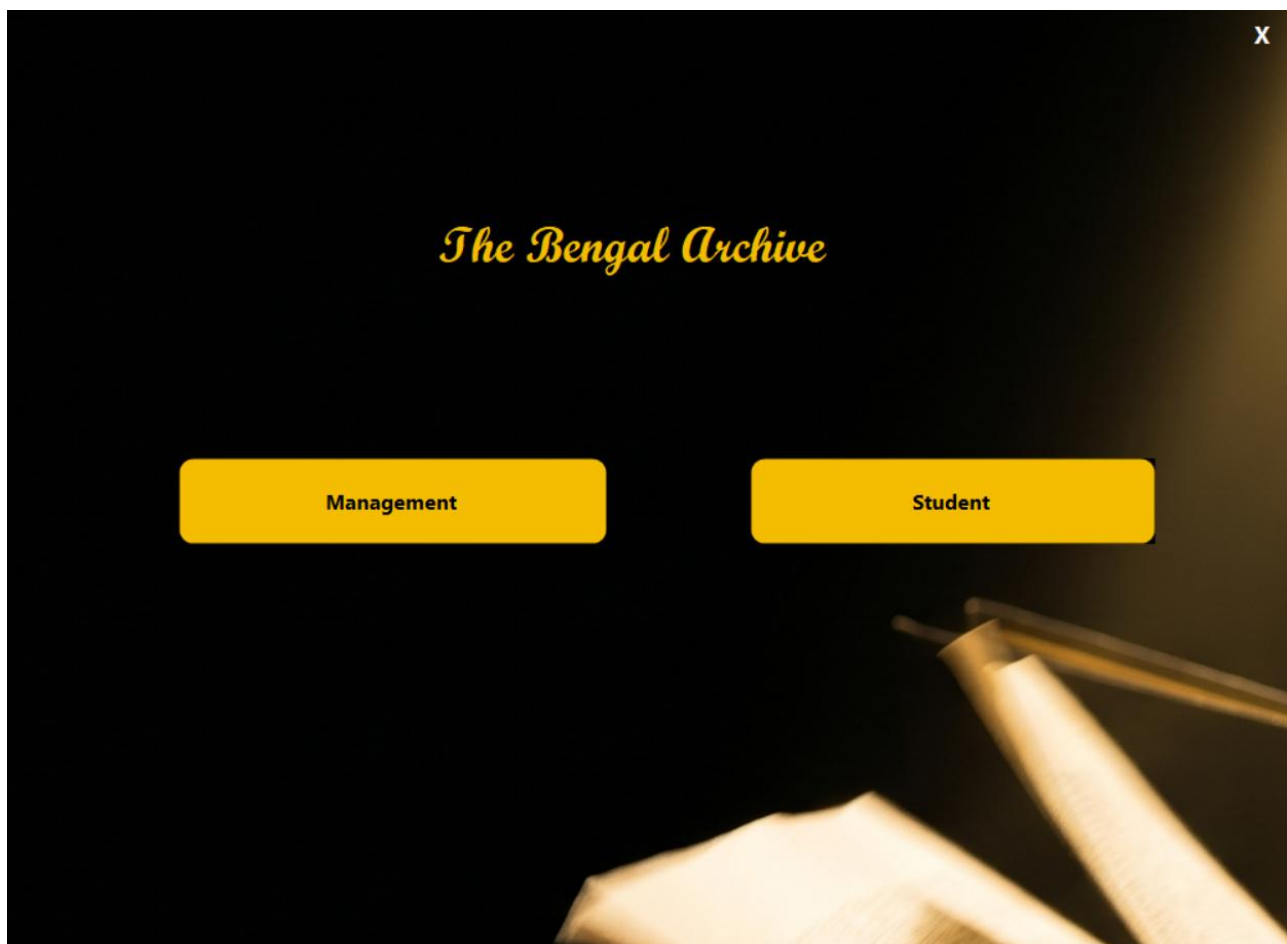
2.2 User 2: Student

- **Dynamic Viewing:** Students can view the live catalog of all available books.
- **Wishlist System:** Students can add books to a personal wishlist.
- **Logic Gate Restrictions:** The system enforces a strict 3-book limit per student and actively prevents duplicate wishlist entries.
- **Wishlist Management:** Students can toggle between viewing the global library or their personal wishlist, and remove books to free up their wishlist capacity.

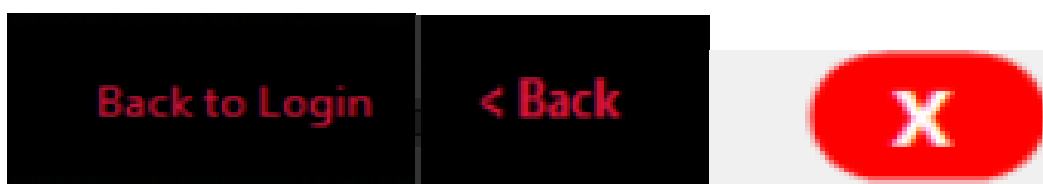
2.3 User 3: Guest / New User

- **System Registration:** Unauthenticated users can access the registration portal to create a new account.
- **Role Selection:** Users must define their role (Management or Student) upon creation, which dictates their future access rights.

Welcome Screen



Animated Buttons



Login and Register Page

< Back

X

Welcome Learner!

Enter your Full Name

Enter Password

REGISTER

Back to Login

< Back

X

Welcome Back Learner!

Fahim

.....|

LOGIN

New here? Create an account

Management Dashboard

Learners

Fine

Manage Books

+ Add New Book

- Remove Book

The Bengal Archive

Logout

X

1	C# by Shimu	Unknown	General	1	1	5/17/2026 11:21 PM
3	ABCD Programmi...	Unknown	General	1	1	5/18/2026 6:42 PM
4	The Nagasaki	Unknown	General	1	1	5/18/2026 6:44 PM
5	চলো যুগাই	Unknown	General	1	1	5/18/2026 6:45 PM
6	rgfaefaw	Unknown	General	1	1	5/18/2026 7:31 PM

Add New Book

Enter Book Name:

ADD BOOK

Remove Book

Select Book to Remove:

ABCD Programming

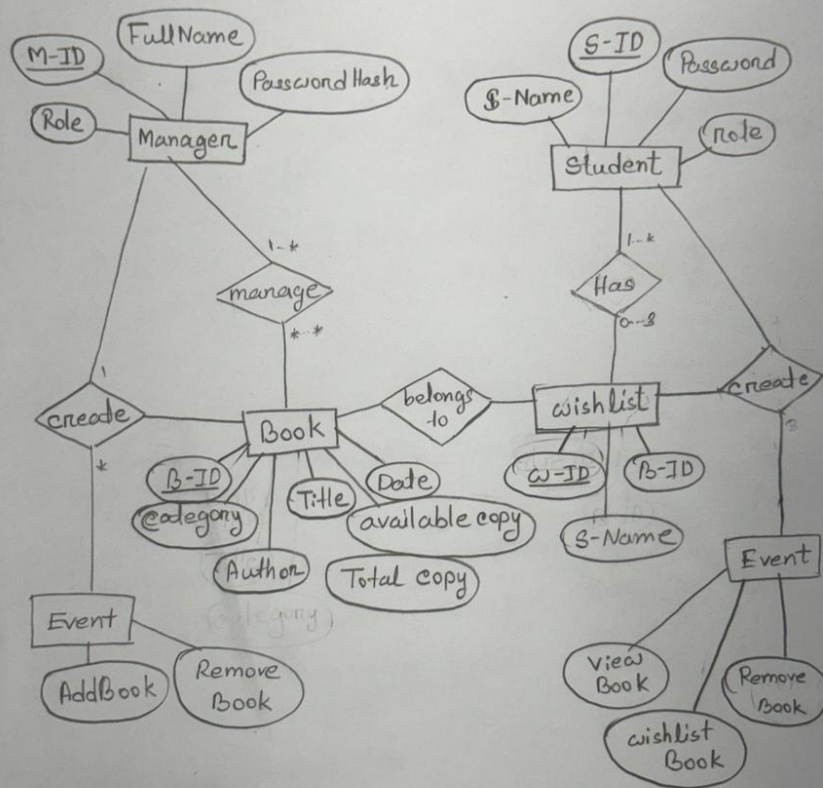
X REMOVE BOOK

Student Dashboard

						X
1	C# by Shimu	Unknown	General	1	1	5/17/2026 1...
3	ABCD Progr...	Unknown	General	1	1	5/18/2026 6:...
4	The Nagasaki	Unknown	General	1	1	5/18/2026 6:...
5	চলো ঘুমাই	Unknown	General	1	1	5/18/2026 6:...
6	rgfaefaw	Unknown	General	1	1	5/18/2026 7:...

ER Diagram

ER-Diagram



Normalization:

① Relation: Many Managers manage many books.

UNF:

M-ID, Role, FullName, PasswordHash

B-ID, Category, Author, Title, Date, Total copy, available copy

1NF:

① M-ID, Role, FullName, PasswordHash

② B-ID, category, Author, Title, Date, Total copy, available copy

2NF:

① M-ID, Role, FullName, PasswordHash

② B-ID, category, Author, Title, Date, Total copy, available copy, M-ID

3NF:

① M-ID, Role, FullName, PasswordHash

② B-ID, category, Author, Title, Date, Total copy, available copy, M-ID

② Relation Many student has 3 wishlist

UNF:

S-ID, Sname, Password, role

W-ID, B-ID, S-name

1NF: ① S-ID, Sname, Password, role

② W-ID, B-ID, S-name

2NF: ① S-ID, S-name, Password, role, W-ID

② W-ID, B-ID, S-name, S-ID

3NF: ① S-ID, S-name, Password, role, W-ID

② W-ID, B-ID, S-name, S-ID

③ Relation:

Wishlist belongs to many Books

2NF: ① W-ID, B-ID, S-Name

② B-ID, category, Author, Title, Date, available copy,
Total copy, W-ID

Finalization:

① M-ID, Role, FullName, PasswordHash

② B-ID, category, Author, Title, Date, Total copy, available
copy, M-ID

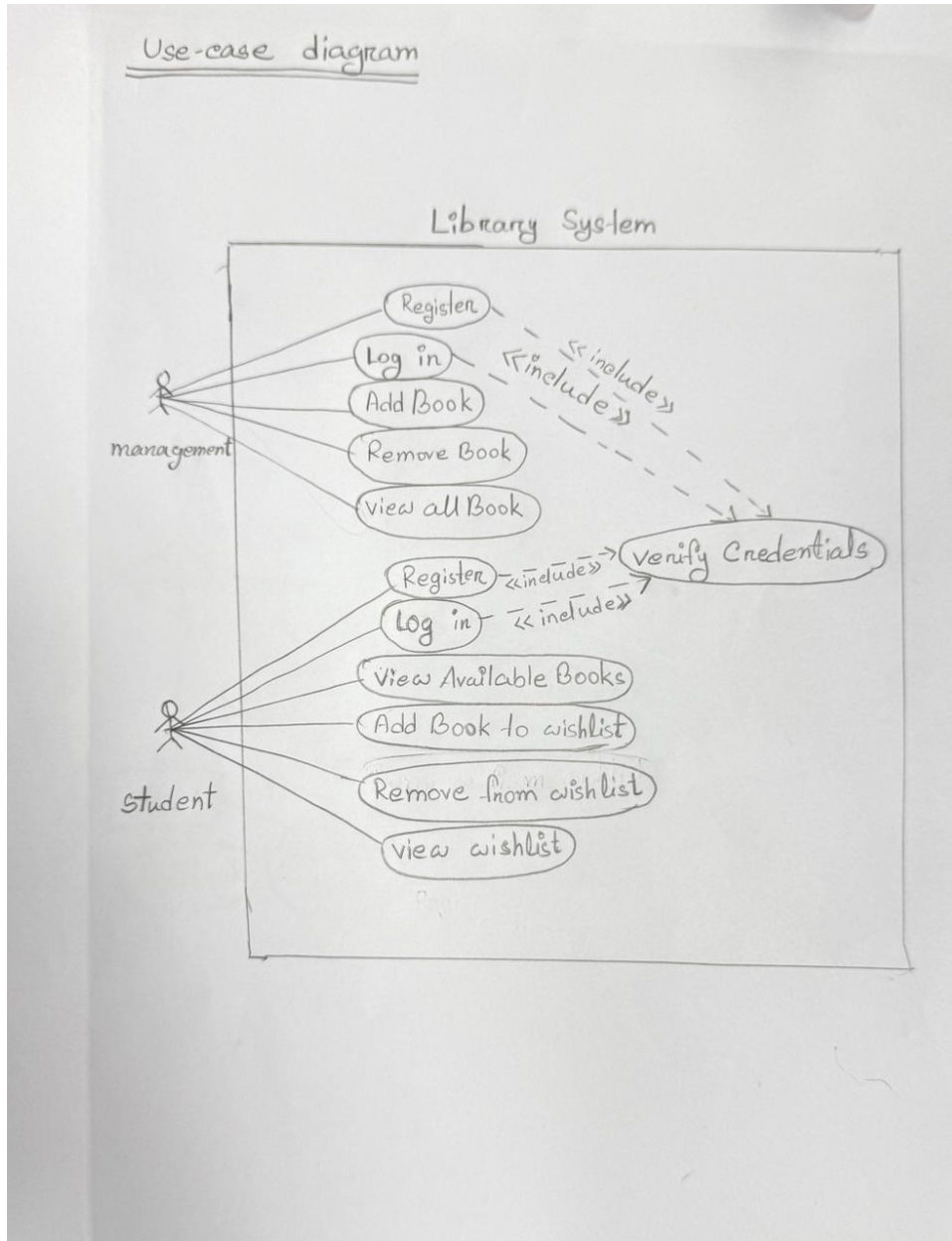
③ S-ID, S-name, password, role, W-ID

④ W-ID, B-ID, S-name

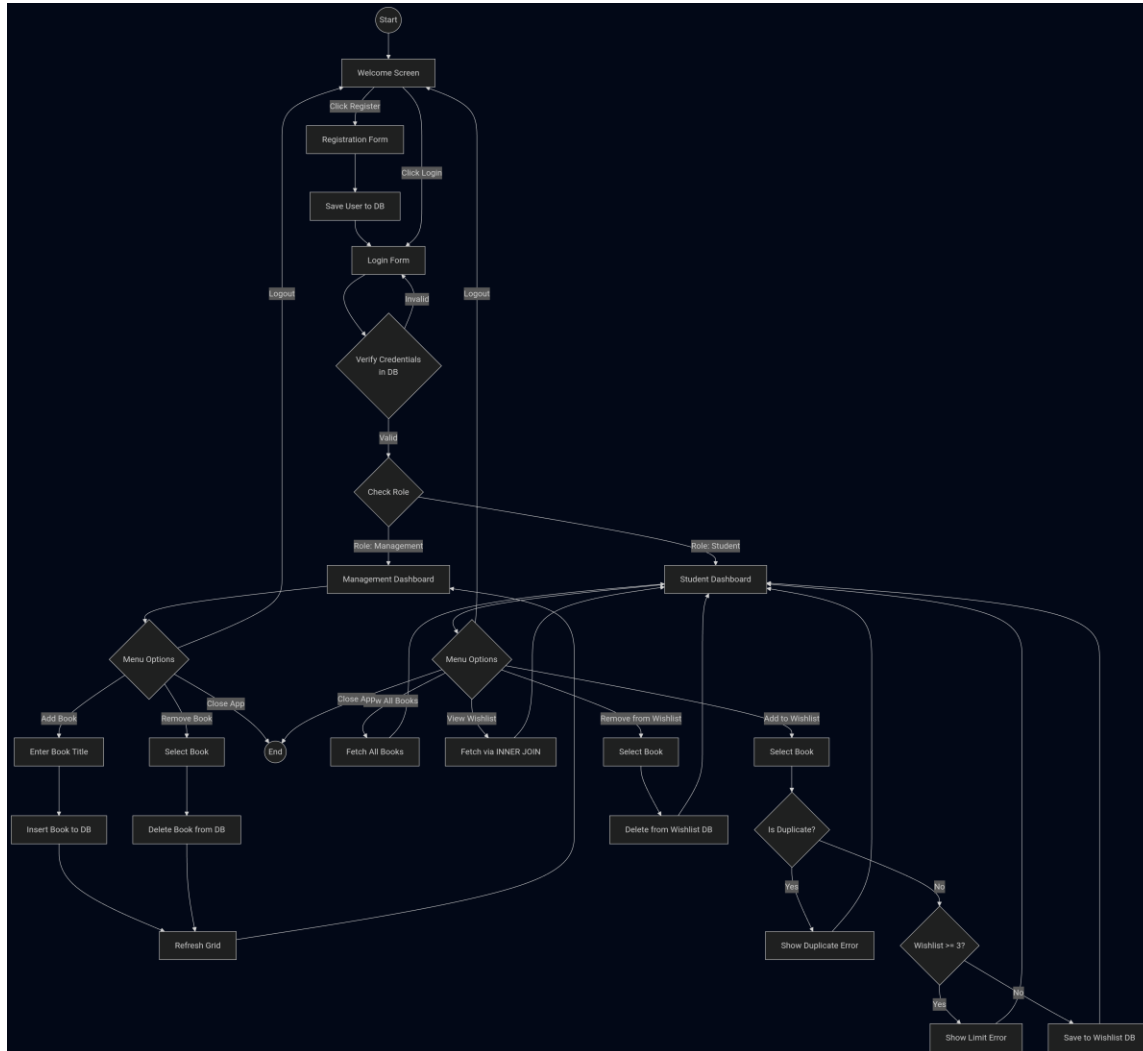
⑤ B-ID, category, Author, Title, Date, available copy,
Total copy, W-ID

UML Diagram

Use case diagram



Activity Diagram



5. CONCLUSION

The development of this Library Management System successfully demonstrated the practical application of Object-Oriented Programming concepts, secure data handling, and multi-tier software architecture. By decoupling the UI from the database logic via the BLL and DAL, the application remains highly scalable and easy to debug. The implementation of robust logic gates, such as preventing duplicate wishlist entries and managing user capacities, highlights a strong focus on defensive programming and user experience. Ultimately, the project serves as a comprehensive, real-world example of bridging C# application logic with relational database management.